Krishna Subedi

Full-Stack Engineer — Backend Systems & Web Platforms

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Experience

Neryva Jul 2025 – Present

Member of Technical Staff · Remote

- Architected a multi-tenant NestJS/Fastify control plane with in-engine OIDC, step-up MFA, transaction-local RLS, AES-256-GCM envelope encryption for BYOK, and provider-neutral PostgreSQL/MongoDB persistence with parity checks.
- Built a Buf-managed ConnectRPC/Protobuf contract with breaking-change CI. Added a transactional SKIP LOCKED outbox, two-tier request-fingerprinted Redis/SQL idempotency, and DNS-pinned SSRF controls blocking private and metadata IPs
- Built Agent Studio's React console with TanStack Router/Query and Zustand. Implemented a configuration-first builder with draft autosave, HTTP 412 conflict handling, dedicated create/edit routes, and session-scoped SSE token streaming.

Smaitic Labs Apr 2024 – Jul 2025

Full-stack Developer

- Designed Vanidya's three-tier admin model (main, state, and seller) with hierarchical RBAC and tenant-level data isolation. Built a dashboard for each tier. Region and seller onboarding required data changes only, with no schema or code changes.
- Built Vanidya's multi-tenant Spring Boot backend with ZIP-code-keyed delivery routing and coverage. Implemented an automated seller commission and fee engine, replacing manual per-order fee calculation.
- Shipped real-time order tracking across native Android (Kotlin) and offline-first PWA clients, synchronizing order status across buyer and seller views.

Nepal Oil Corporation Limited Oct 2023 – Mar 2024

Software Engineer

- Designed a dynamic time-series forecasting system for regional diesel, petrol, and aviation fuel demand, modeling shock surges from strikes and festivals to optimize depot inventory allocation and prevent consumer-side pump stockouts.
- Engineered automated Python and SQL pipelines consolidating national depot and tanker movements into live operational dashboards, replacing manual end-of-day reporting with continuous dispatch visibility.

Projects

Krishi Vaidya: AI-Powered Crop Disease Diagnosis Apr – Jul 2026

Personal Project · Full-Stack

- Built an end-to-end crop disease diagnosis platform: an Expo Router mobile app with on-device TFLite/OpenCV inference for offline, zero-latency diagnosis in low-connectivity fields, and an Express.js API.
- Fine-tuned Qwen2.5-VL-3B via LoRA and trained a YOLO-based disease detector for multi-stage crop diagnosis, integrated with a Gemma-powered advisory pipeline and shipped to production iOS/Android apps.

Ecommerce Platform: Multi-Seller Marketplace Apr – Jun 2025

Personal Project · Full-Stack

- Built a multi-seller marketplace backend with five-role RBAC (customer, individual seller, vendor staff, vendor admin, superadmin) and vendor-scoped permission checks, alongside a multi-step Stripe checkout supporting saved payment methods, shipping, tax, and discount rules.
- Architected and delivered a multi-seller storefront and role-based admin portal, designing a variant-aware product catalog and client-side session-to-account cart reconciliation.

Education

Bachelor of Engineering in Mechanical Engineering 2019 – 2023

Tribhuvan University · Licensed PE

Skills

Languages: C/C++, Python, TypeScript, Java, Kotlin, SQL, JavaScript · **AI/ML:** Generative AI/LLMs, RAG Pipelines, LLM Evaluation, Time-Series Forecasting · **Web/Backend:** Spring Boot, Node.js (Express 5), FastAPI, Django, React 19, Redux · **Data/Cloud:** PostgreSQL, MySQL, MongoDB, Redis, Docker, Kubernetes, Jenkins, AWS · **Architecture/Practices:** Microservices, Multi-Tenancy, CI/CD, RBAC, REST APIs, Linux, Jest, PWA